Assignment 4

CSE260: Digital Logic Design

Department of Computer Science and Engineering

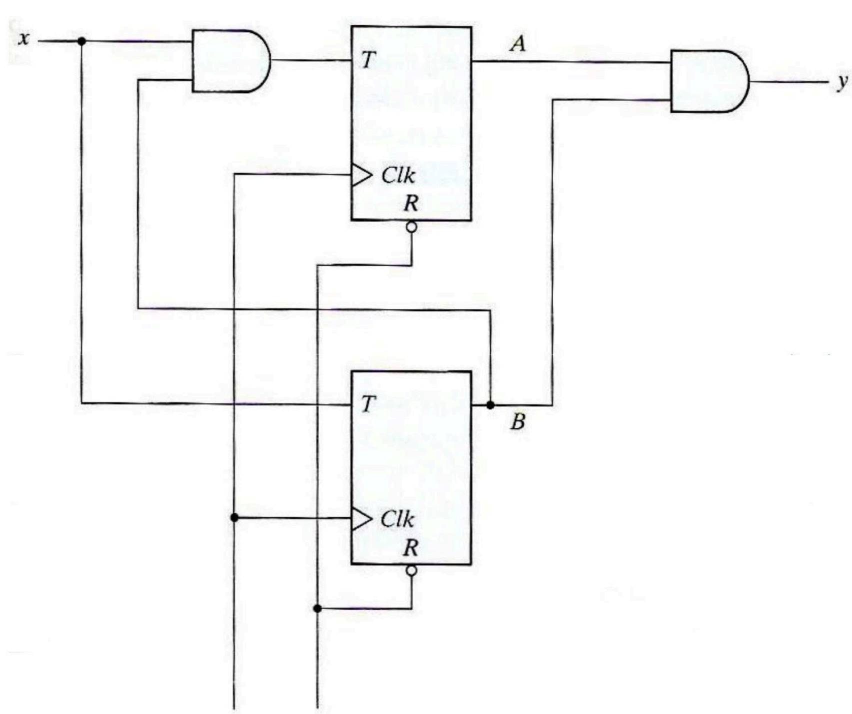
Semester : Spring26 **Marks: 5**

Graded (5 marks)

1. Short Questions:
    a. Explain the difference between a latch and a flip-flop. [0.5]
    b. Describe the working of a JK flip-flop and explain how it overcomes the undefined state of an SR flip-flop. [0.5]
    c. What is the importance of the clock signal in flip-flops? What happens when we use the same clock for multiple flip-flops [0.5]
    d. Why do we need state diagrams to understand sequential circuits? [0.5]

2. Implement the following up/down counter using SR FF: 2 ↔ 4 ↔ 5 ↔ 6 ↔ 7 ↔ 2
   Note: Represent each number using ABC variables. Consider the selection bit as: M=0 (up) and M=1 (down). The order of the variables must be **MABC**. [1]

3. Convert the following circuit diagram to state diagram: [1]



4. You are a BRAC University student whose focus and priorities shift constantly throughout the semester. The semester began with full motivation as you attended every lecture, completed assignments ahead of deadlines, and even visited the **Library** regularly to get ahead on readings. However, as midterms approached, the stress became overwhelming and you found yourself spending most of your time in the **Cafeteria**, stress-eating with friends and avoiding responsibilities. After receiving a disappointing midterm result, guilt kicked in and you rushed back to the **Library**, determined to turn things around. This time, a classmate suggested a change of scenery, and you migrated to the **Department** lobby to study in a different environment. The productivity was short-lived however, as friends spotted you and convinced you to head up to the **Rooftop** to take a long overdue break.

   But the smell of food drifting up from below was too hard to resist, and you soon found yourself back in the **Cafeteria**. After eating, a sense of responsibility returned and you made your way back to the **Library** to refocus. This time you managed to stay productive for a while, but an announcement about a group project meeting sent you over to the **Rooftop** once again. But this gave you an opportunity to take a break as well.. The fresh air and city view were refreshing, but with finals just around the corner, you dragged yourself back to the **Library** one final time, determined to finish the semester strong.

   Design and draw the complete circuit diagram representing your semester journey using **JK Flip-Flop(s)**. In case of missing transition(s), use **Don't Cares** for the next state.